

NATURAL LANGUAGE UNDERSTANDING (CSCI 6515)

PROJECT REPORT

SEMANTIC DRIFT

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GitHub code repository: <https://github.com/vishrutharavi/semantic-drift>

Overview of the Task

Language is not static; it evolves continuously over time as new technologies, cultural shifts, and social interactions introduce new meanings and usages for existing words. This phenomenon, known as semantic drift, refers to the gradual or abrupt change in the meaning of words across different time periods and contexts. The goal of this project is to detect, quantify, and analyze semantic drift by leveraging large-scale textual data from two distinct sources: historical corpora (news articles) representing earlier or traditional meanings, and modern social media data (Reddit) representing emerging and evolving meanings. By comparing these domains, the project aims to capture how language transitions from formal, stable usage to dynamic, community-driven evolution.

The project involves multiple stages. First, raw textual data from both sources is cleaned, standardized, and organized into a temporal structure to enable time-based analysis. Next, relevant contexts for selected words are extracted and transformed into numerical representations using contextual embedding models, which capture semantic meaning based on usage. These embeddings allow comparison of word meanings across time, and clustering techniques are applied to group similar contexts into distinct semantic usages. A drift score is then computed using embedding-based similarity measures to quantify the degree of change, and visualization methods are used to make these shifts observable. Additionally, a Large Language Model (LLM) is incorporated only to generate human-readable explanations of the observed semantic changes, while the actual drift detection relies on embedding-based similarity thresholds. This separation ensures robustness, as LLMs are prone to overestimating semantic drift. Overall, the task focuses not just on identifying whether a word has changed meaning, but on understanding how it has evolved, to what extent, and in which contexts.

Motivation

Words that once had stable and well-defined meanings are now frequently reinterpreted, repurposed, or extended to new contexts, especially in fast-evolving digital environments. Terms like “*viral*” or “*cloud*” illustrate how meanings can shift significantly across domains and over time. This rapid evolution poses a major challenge for natural language processing (NLP) systems, which often assume that language is relatively stable. Traditional linguistic approaches rely on curated historical texts and manual analysis, which, while insightful, do not scale to the vast and continuously growing volumes of modern data. As a result, NLP models may become outdated or less accurate when they encounter newly evolved meanings.

This project is motivated by the need to automatically detect and quantify semantic drift in real-world data. By identifying when and how word meanings change, the system can help determine when models need retraining, improving efficiency while reducing unnecessary computational cost. Additionally, by bridging the gap between formal language (news) and informal, rapidly evolving language (social media), the project captures both long-term stability and short-term innovation in language. Ultimately, the goal is to develop a scalable, data-driven approach to understanding language evolution and to support the creation of more adaptive and robust NLP systems.

Contributions

- Developed an end-to-end pipeline for semantic drift analysis, covering data preprocessing, context extraction, embedding generation, clustering, and drift computation.
- Leveraged multi-source data (news + Reddit) to capture both traditional and evolving word meanings across domains.
- Implemented automated candidate word selection, enabling scalable analysis instead of manual word picking.
- Used contextual embeddings (Sentence Transformers) to represent word meanings based on usage rather than static definitions.
- Applied clustering techniques (KMeans) to identify distinct semantic usages of words.
- Designed a drift score using embedding similarity to quantitatively measure semantic change.
- Built visualization components (UMAP projections) to observe semantic shifts and cluster separations.

- Integrated an LLM-based explanation module (LLaMA 3.1 via Ollama and LangChain) to generate human-readable interpretations of semantic changes, improving interpretability while keeping drift detection strictly embedding-based.
- Maintained a clear separation between drift detection (embedding-based) and explanation (LLM-based) to ensure robustness.
- Provided a scalable framework that can be extended to analyze large vocabularies and support real-world NLP applications.

Literature Review

- Hamilton et al. (2016)
 - Introduced diachronic word embeddings to study semantic change over time
 - Proposed quantitative metrics to measure how word meanings shift across historical corpora
- Kutuzov et al. (2018)
 - Provided a comprehensive survey of computational approaches for lexical semantic change detection
 - Covered methods using neural embeddings and temporal text data
- Garg et al. (2018)
 - Demonstrated that contextual word embeddings can capture long-term linguistic shifts
 - Showed how embeddings can also reflect societal and cultural changes over time

Methods

1. Data Collection and Preprocessing

The project uses two complementary data sources: **news articles** to represent earlier, more stable meanings and **Reddit posts** to capture modern, evolving usage. The text is cleaned by removing noise such as URLs, special characters, and formatting inconsistencies. It is then organized into a **temporal structure (monthly JSONL files)** so that usage can be analyzed over time.

Motivation: Proper preprocessing ensures consistency across datasets and allows meaningful temporal comparisons, which are essential for studying how language evolves.

2. Candidate Word Selection

Instead of manually choosing words, the system identifies candidate words automatically based on their **frequency and presence across both datasets**. This ensures that only words with sufficient contextual data are selected for analysis.

The system first extracts a ranked list of candidate words that may show semantic change. Instead of manually selecting only a few words, the pipeline computes drift-related statistics for many words and stores the **top 100 candidate words** for later analysis. This makes the project more scalable because the system can analyze multiple words automatically rather than depending on fixed examples.

Motivation: Automation improves scalability and reduces bias, enabling the system to analyze a larger vocabulary and focus on words that are more likely to exhibit semantic variation.

3. Context Extraction

For each selected word, sentences containing that word are extracted from both datasets. These are then divided into **early contexts (from news)** and **late contexts (from Reddit)**. Each sentence serves as an instance of how the word is used in a specific context.

For each of the top 100 candidate words, the system extracts context examples from both news and Reddit data. The news contexts represent older or more formal usage, while Reddit contexts represent newer and informal usage. These contexts are stored so that the user can later request any number of results, such as the **top 10, 20, 30, 60, or 70 drifting words**, without rerunning the entire extraction process.

Motivation: Word meaning is defined by its usage in context. Extracting multiple contextual examples allows the system to capture different semantic interpretations of the same word.

4. Embedding Generation (Sentence Transformers)

Each extracted sentence is converted into a numerical vector using **Sentence-BERT (all-MiniLM-L6-v2)**. These embeddings capture the semantic meaning of the sentence based on context, rather than just the individual word.

Motivation: Contextual embeddings are essential for detecting semantic drift because they reflect how words are used in real sentences. The chosen model is both efficient and effective for large-scale processing.

5. Clustering (K-Means)

The sentence embeddings are grouped using **K-Means clustering**, where each cluster represents a distinct semantic usage of the word. Typically, a small number of clusters (2) is used to separate major meanings, such as traditional vs modern usage.

Motivation: K-Means provides a simple and interpretable way to identify different meanings. It works well with dense embeddings and helps visualize how contexts group into semantic categories.

6. Drift Score Computation

Semantic drift is quantified by comparing early and late embeddings using **cosine similarity**. A drift score is computed based on how far apart these two sets of embeddings are in the vector space.

Motivation: This provides a clear, numerical measure of semantic change. Cosine similarity is widely used for comparing embeddings and allows consistent comparison across different words.

7. Visualization (UMAP)

High-dimensional embeddings are projected into 2D space using **UMAP**, enabling visualization of clusters and their separation. This helps illustrate how meanings shift and whether clusters overlap or diverge.

Motivation: Visualization makes abstract embedding relationships easier to interpret and supports qualitative analysis of semantic drift.

8. Top-N Output

After the top 100 candidate words and their contexts are prepared, the system supports flexible user queries. For example, if a user asks for the top 30 drifting words, the pipeline filters and ranks the saved candidate list by drift score and returns the top 30 available words. The same process works for top 10, 20, 60, or 70 words. This makes the system interactive and efficient because the heavy preprocessing and context extraction are already completed.

Motivation: Precomputing candidate words and their contexts avoids repeated expensive processing and enables fast, on-demand analysis. It also allows users to explore semantic drift at different scales (top 10 vs top 100) without rerunning the entire pipeline, improving both efficiency and usability.

9. Filtering and Quality Control

The pipeline applies multiple filtering steps to remove weak or noisy candidates, including misspellings, **offensive or inappropriate (bad) words**, and proper nouns that do not represent true semantic drift. It also ensures that each word has a sufficient number of context examples from both datasets before being included in the analysis. This process improves the reliability of the results by focusing only on words with strong and meaningful contextual signals, reducing noise and preventing misleading interpretations.

Motivation:

Raw text data, especially from social media, is often noisy and inconsistent. It may contain spelling errors, named entities, and inappropriate terms that do not reflect meaningful language evolution. Without filtering, the model may detect false semantic changes driven by such noise rather than genuine shifts in meaning. Applying quality control ensures that the analysis is based on **valid linguistic patterns**, leading to more accurate and trustworthy semantic drift detection.

10. LLM-Based Explanation Module

A Large Language Model (LLM) is used to generate human-readable explanations by analyzing representative examples from different clusters. In this project, the **LLaMA 3.1 (8B) model is accessed using the Ollama framework via LangChain (ChatOllama)**, with a low temperature setting (0.2) to ensure more consistent and focused outputs. The model takes selected old and new context examples as input and describes how the meaning of a word has changed over time.

Motivation: While embedding-based methods provide strong quantitative evidence of semantic change, they are not directly interpretable. The LLM adds an explanation layer, making the results easier to understand for humans. At the same time, drift detection is kept strictly embedding-based to ensure robustness, as LLMs can sometimes overestimate or hallucinate semantic differences.

Results and Evaluation

The effectiveness of the proposed semantic drift detection pipeline is evaluated using a combination of quantitative metrics (embedding similarity, drift score) and qualitative analysis (clustering behavior and context interpretation). Since there is no labeled ground truth for semantic drift, the evaluation focuses on measuring relative semantic change and validating it through contextual examples and visualizations.

Evaluation Metrics and Motivation

Cosine Similarity (Embedding Similarity)

Measures how similar early (news) and late (Reddit) contexts are in the embedding space.

Motivation: Widely used for comparing semantic representations; provides a direct measure of how much meaning has changed.

Drift Score

Quantifies the degree of semantic change between two time periods.

Motivation: Converts similarity into an interpretable “change” metric, allowing easy comparison across words.

<pre>old_contexts = news_contexts_for_current_word new_contexts = reddit_contexts_for_current_word Analyzing: stream Loading existing news contexts for 'stream'... Loading existing reddit contexts for 'stream'... Prepared contexts for 'stream': News contexts: 1500 Reddit contexts: 2576 Drift score: 0.28321475</pre>	<pre>Analyzing: viral Loading existing news contexts for 'viral'... Loading existing reddit contexts for 'viral'... Prepared contexts for 'viral': News contexts: 1500 Reddit contexts: 2787 Drift score: 0.28926712</pre>
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Fig 1 & 2 . Drift scores for “stream” and “viral”

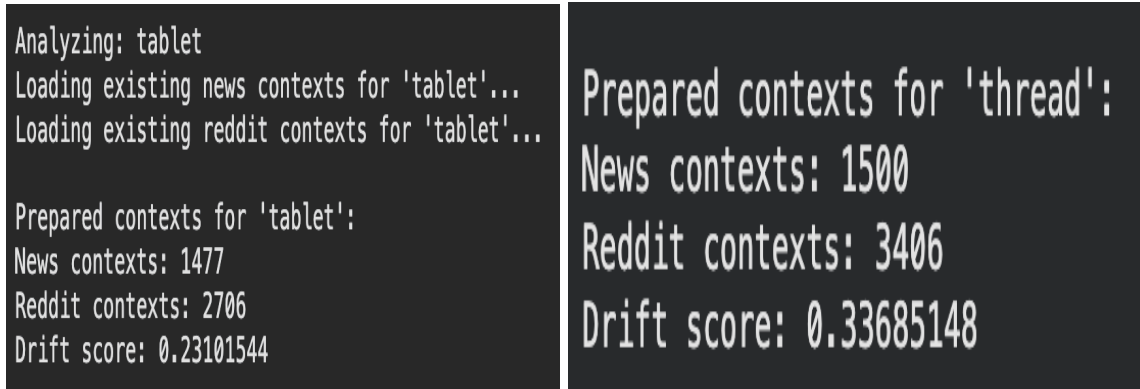


Fig 3 & 4 . Drift scores for “tablet” and “thread”

Clustering (K-Means)

Groups contexts into distinct semantic usages.

Motivation: Helps identify whether multiple meanings exist and whether they are clearly separable.

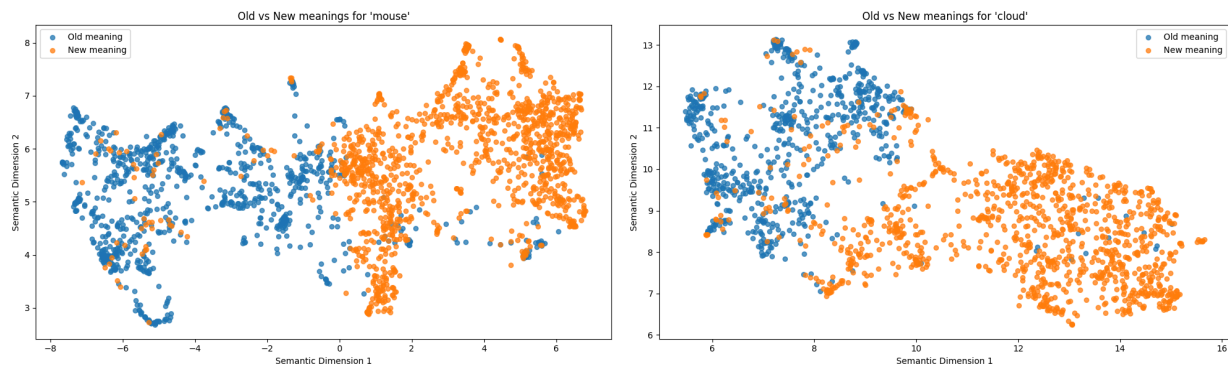


Fig 5 & 6

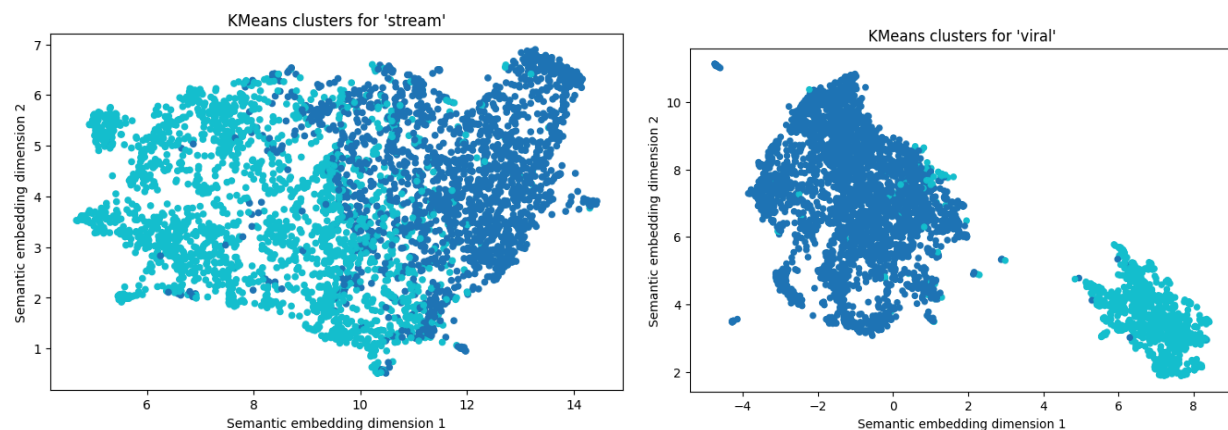


Fig 7 & 8

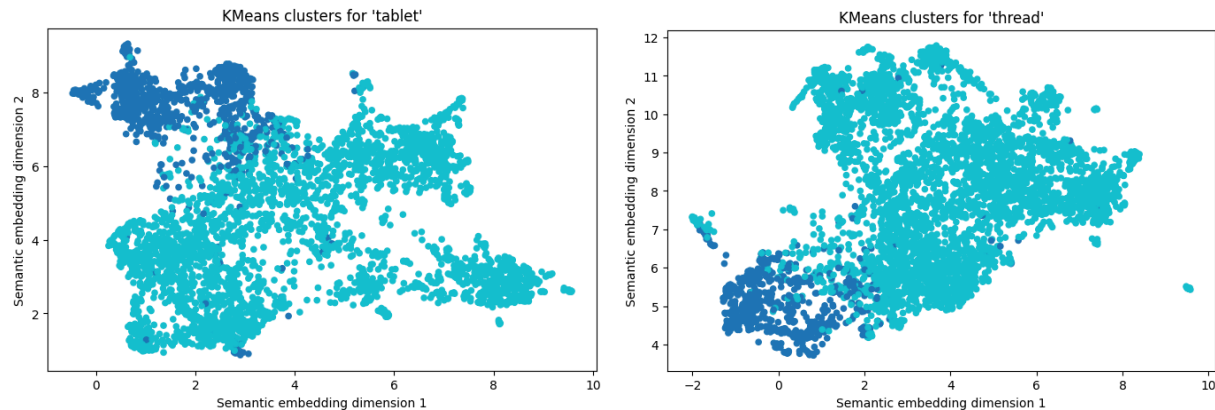


Fig 9 & 10

Fig 5, 6, 7, 8, 9 and 10. K-Means Clustering of Contextual Embeddings for “mouse”, “cloud”, “stream”, “viral”, “tablet” and “thread”

Cluster Examples:

1. “mouse”

Cluster: 0

- At the end of his speech, Ban put his hand on the mouse of a laptop computer together with Jones, of Google; Carlos Dominguez, a Cisco Senior Vice President; and Kemal Dervis, U.N. Development Program Administrator.
- But end-users have traditionally downloaded megabytes of information from the web, while uploading only kilobytes of key strokes and mouse clicks.
- Finally, when Google shows you its suggested driving route on the map, you can drag that line onto another road with your mouse.
- We then watch a video of a driver's-eye view of a road (somewhere in Reading, according to the signposts) and click our mouse when we see a hazard.
- Internet Explorer rather redundantly displays the "alt" tag when the mouse is hovered over the image; Firefox doesn't bother because no alternative is needed and there is a 'title' tag for captions.
- Now businesses are increasingly using networking tools such as LinkedIn and Facebook to identify those people with the click of a mouse.
- The Air Mouse is a rechargeable cordless mouse that can control a windows PC by being moved through the air or on a flat surface.
- They're tracking where your mouse is on the page, what you put in your shopping cart, what you don't buy.
- Review: Logitech VX Nano Cordless Laser Mouse praise-worthy
- If you miss the scroll bar found on almost every Windows laptops, the two-finger scroll option works well (run two fingers down the touch pad, and it scrolls like a mouse wheel).

Cluster: 1

- An Oklahoma State University lab in Stillwater in December could not account for a dead mouse inoculated with bacteria that causes joint pain, weakness, lymph node swelling and pneumonia.
- The only time I thought "what's she got me into now?" was when I was ice skating, holding hands with Minnie Mouse.

- Across the park, hot dog vendors sought refuge under the palm trees" slowly shifting shadows, their carts painted with slap-dash cartoon characters like a Mickeyesque mouse and a clumsily drawn Pokémon.
- Obama Gobbles Down Mouse _ Made of Candy
- Importantly, in the presence of HIV, ZFN-modified CD4 T-cells also preferentially expanded in a mouse model.
- Mouse virus link to breast cancer
- After 40 minutes or so, the upstairs contingent was escorted to a smaller room below, where Mr. Roccoli, dressed in white pants and a grubby Mickey Mouse T-shirt (unbecoming indeed), performed a solo to more extracts from interviews and some uncredited songs.
- In one neighborhood of Rawalpindi, police played a cat-and-mouse game with the protesters.
- Christmas cracker contains dead mouse
- She may be 79 years old, but Minnie Mouse inspires very up to the minute designers.

2. "viral"

Cluster: 0

- Doing the 'Aunt Jackie': A Harlem-filmed rap video goes viral on MySpace.
- Embarrassing moments go viral too, as when Miss South Carolina Teen USA Lauren Caitlin Upton was asked during the televised pageant to answer the question "Recent polls have shown a fifth of Americans can't locate the U.S. on a world map.
- Our launch program is built on a Viral Marketing and college grassroots strategy to capitalize on the inherent appeal that we believe ThePeeq.com will have with their audience.
- Viral events are unpredictable.
- By some estimates, the Star Wars kid is the most popular viral video ever, and viewed over one billion times.
- A writer pays for breaking the strike in this "viral" video "Viral" video
- Hupo.TV, recognized recently by PC World China as one of the top 10 video sites in China, is known for its original and contagious/viral content that keeps the consumers coming back for more.
- There's no evidence that Adblock Plus or similar products are about to go viral.
- Funnyordie's Adam McKay has directed Ferrell in \$100 million movies like "Anchorman," but he was just as excited when "The Landlord" went viral because Pearl is his daughter.
- Launched by Sean Parker (a Zuckerberg pal and partner in the Founders Fund, which owns a stake in Facebook) and Joe Green, the application allows Facebook users to tap into the latent viral power of their social networks for good causes.
- Add in the viral nature of the contest and it proved to be a winning formula.
- Loiacono, a Warcraft fan, was looking to create some viral buzz for his start-up, a social networking site called Voig aimed at bringing video game fans together.
- If you're browsing the Web on company time looking for the latest in viral videos, you're not alone.
- Companies everywhere are monitoring blogs and other online discussions for feedback on their brands and providing them with information about coming products, as well as placing so-called viral advertisements on video-sharing sites.
- The four artists left standing will battle it out for the (not contagious) "Viral Woodie."

Cluster: 1

- The investigational vaccine used in both ... studies cannot cause HIV infection because it contains only synthetically produced snippets of viral material.
- A viral heart infection called cardiomyopathy is to blame, People magazine reports.

- Two other wards at the hospital have closed to new admissions because three patients have suspected viral gastroenteritis.
- Using the company's proprietary virus-like particle (VLP) and Novasome (R) adjuvant technologies, Novavax is developing vaccines to protect against H5N1 pandemic influenza, seasonal flu and other viral diseases.
- treatment options (which could include surgery, radiation, chemotherapy, transplants, clinical trials, biological, hormonal, viral and complementary therapies and alternative therapies),
- Idenix Pharmaceuticals, Inc., headquartered in Cambridge, Massachusetts, is a biopharmaceutical company engaged in the discovery, development and commercialization of drugs for the treatment of human viral and other infectious diseases.
- Meningococcal disease often begins with symptoms that can be mistaken for common viral illnesses, such as the flu.
- The animal which tested positive had one of the highest viral counts ever observed in a red squirrel, said Dr Shuttleworth, and it was likely that other wild red squirrels would have been exposed to the virus.
- Ardea Biosciences is focused on the discovery and development of small-molecule therapeutics for the treatment of viral diseases, cancer and inflammatory diseases.
- After administration, tumor-specific viral colonization enables the immune system to recognize tumors as foreign elements and activate defense mechanisms to fight them.
- However, colds and flu are viral, not bacterial, so antibacterial soap doesn't help fight the illnesses, says Sampathkumar.
- AdHoc (including Viral)
- Hilary Benn, the environment secretary, told parliament Tuesday that officials were doing their "darnedest" to ensure the disease did not spread, and said the anti-viral drug Tamiflu had been given to all those who were involved in the poultry cull.
- Newton is battling cardiomyopathy, a viral infection of the heart, said his spokeswoman Lori Jonas.
- Researchers said it had "significantly" better recovery rates than the anti-viral agent acyclovir, which they claimed "had little benefit."

LLM-based Classification

Labels words as “real semantic change” or “no significant drift.”

Motivation: Adds interpretability and validates whether embedding-based differences correspond to meaningful changes.

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LLM Explanation

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Sidenote:

OLD MEANING: A brief aside or additional comment in writing or speech.

NEW MEANING: An informal remark or anecdote, often used to illustrate a point or share personal experience.

WHY DRIFT: The new examples use "sidenote" to introduce humorous stories or personal anecdotes, whereas the old examples used it for sarcastic comments or tangential information.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: Medium

CODEPENDENT:

OLD MEANING: A relationship where one person is overly dependent on another, often with a power imbalance.

NEW MEANING: An expectation that one's partner must constantly be available and involved in every aspect of their life, with an emphasis on fusion and lack of personal boundaries.

WHY DRIFT: The meaning remains consistent across examples; differences are contextual or stylistic rather than a core meaning change.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: High

UNDERAGED:

OLD MEANING: Refers to a person who is below the legal age of consent or maturity.

NEW MEANING: Refers to a person, typically in their teenage years, who is considered too young for certain activities or situations due to their immaturity or lack of experience.

WHY DRIFT: In the new examples, "underaged" is used to describe someone's perceived immaturity or lack of life experience, rather than just their age. This shift in meaning is evident when the speaker describes themselves as an "underaged male" who would take naked selfies with the intruder in the background.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: Medium

WORD: homeroom

OLD MEANING: A class or period in school where students meet with their teacher for administrative tasks and socialization.

NEW MEANING: A setting where the narrator experienced bullying, teasing, and weight-related issues during his childhood and teenage years.

WHY DRIFT: The meaning remains consistent across examples; differences are contextual or stylistic rather than a core meaning change.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: High

KICKASS:

OLD MEANING: Used to describe something or someone as excellent or impressive.

NEW MEANING: Used to describe something or someone with a violent or aggressive connotation.

WHY DRIFT: The meaning remains consistent across examples; differences are contextual or stylistic rather than a core meaning change.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: High

RESPAWN:

OLD MEANING: To continue fighting or be resurrected in a game after dying.

NEW MEANING: To revive or reappear after a set time period, often with a delay, after dying in a game.

WHY DRIFT: The meaning remains consistent across examples; differences are contextual or stylistic rather than a core meaning change.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: Medium

CATCALLED:

OLD MEANING: To shout insulting or derogatory comments at someone, especially in a public place.

NEW MEANING: To be subjected to unwanted attention or harassment from men, often involving verbal comments that are objectifying or degrading.

WHY DRIFT: The new examples show that "catcalled" is now used to describe the experience of being harassed or objectified by men, rather than just shouting insults.

This shift in meaning is evident in phrases like "I'm okay with how I look for the most part (clothed) and I'm pretty cute most of the time, so I got that cut fat chick aesthetic" and "Men will straight up sympathy talk to you."

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: High

THEISM:

OLD MEANING: Belief in the existence of one or more gods.

NEW MEANING: Unclear from examples.

WHY DRIFT: The meaning remains consistent across examples; differences are contextual or stylistic rather than a core meaning change.

WHEN: First observed in Reddit data (~2015), but unclear exact origin.

CONFIDENCE: Medium

The LLM-generated explanations for the top 8 words provide deeper insight into how semantic drift manifests in real-world usage. From the analyzed examples, it is observed that not all words with noticeable contextual differences correspond to true semantic change.

Among the top words, some cases such as *"catcalled"* and *"underaged"* demonstrate **clear semantic reinterpretation**, where the usage has evolved to reflect more nuanced or socially contextual meanings. For example, *"catcalled"* shifts from simply describing verbal remarks to capturing the broader experience of harassment, indicating a meaningful change in interpretation.

On the other hand, several words such as *"codependent," "kickass,"* and *"respawn"* show **no significant semantic drift**, despite appearing in different contexts. In these cases, the core meaning remains stable, and variations are primarily stylistic or contextual rather than semantic. This highlights an important observation: **contextual variation does not always imply semantic change**.

Additionally, some words such as *"theism"* and *"homeroom"* show **unclear or weak drift signals**, suggesting that insufficient or ambiguous context can limit reliable interpretation. This reinforces the importance of strong contextual data for accurate drift detection.

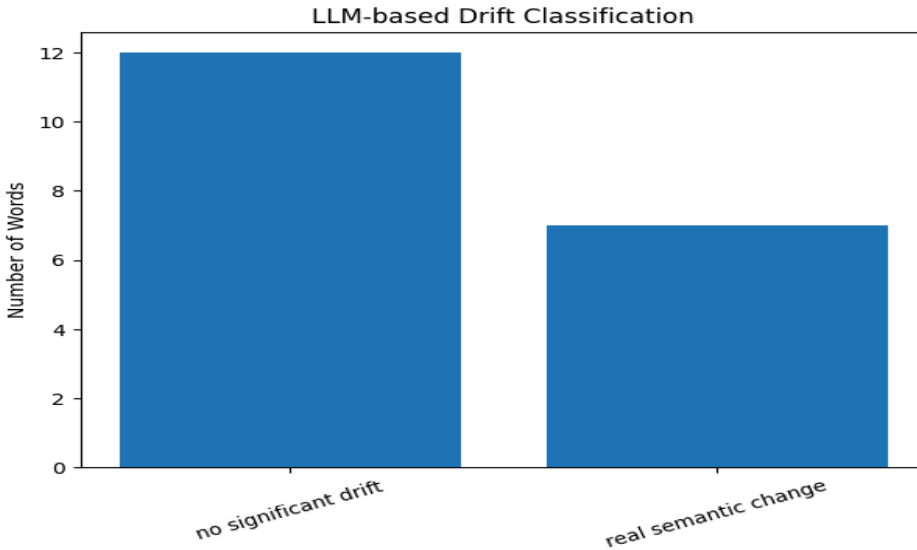


Fig 11. Distribution of Semantic Drift Labels (LLM-Based Classification)

This bar chart shows the distribution of words classified into two categories: “no significant drift” and “real semantic change.” Most of the analyzed words fall under no significant drift, indicating that their meanings have remained relatively stable over time. A smaller portion shows real semantic change, suggesting that only some words undergo meaningful shifts in usage.

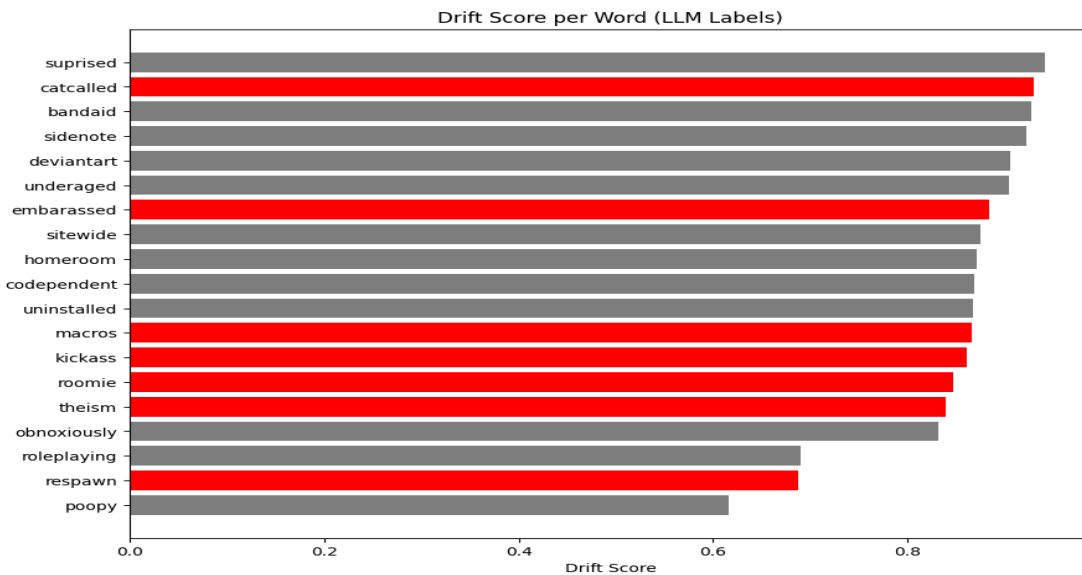


Fig 12. Drift Score Distribution Across Words with LLM-Based Labels

This chart shows the drift score for each word, ranked by how much their meaning has changed. Words highlighted in red represent real semantic change, while gray ones indicate no significant drift. Although many words have high drift scores, only some are classified as

true semantic changes, showing that high numerical drift does not always imply meaningful change.

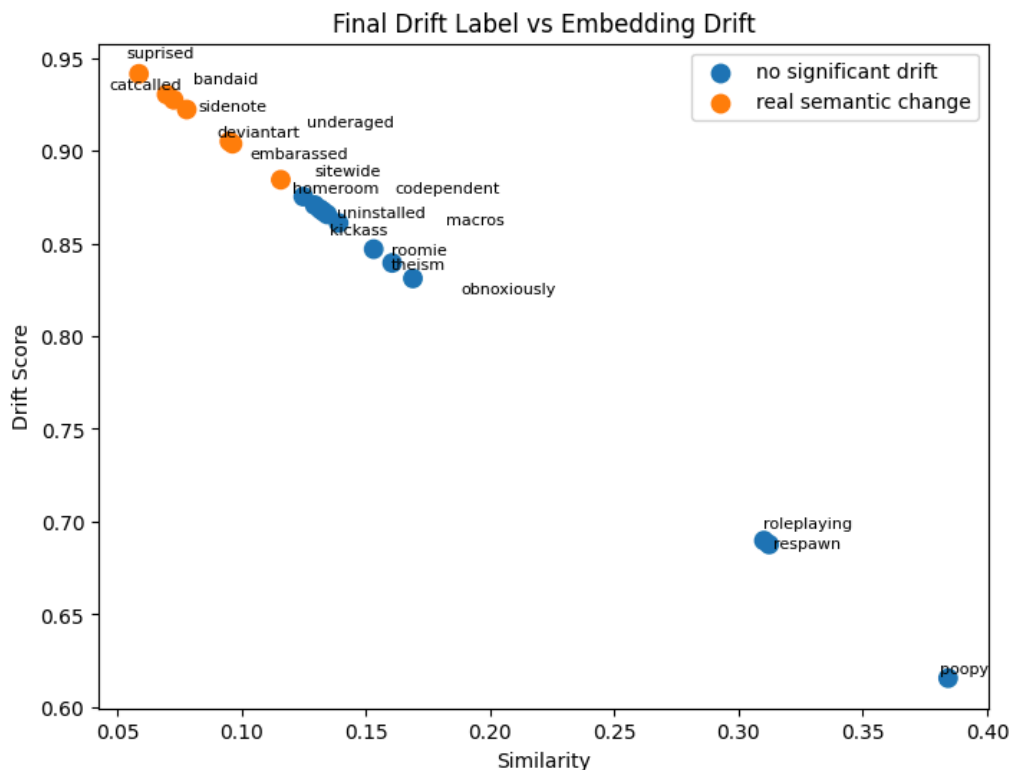


Fig 13. Relationship Between Embedding Similarity and Semantic Drift with Final Classification

This scatter plot shows the relationship between embedding similarity (x-axis) and drift score (y-axis) for each word, along with their final classification. Words with lower similarity and higher drift scores are mostly labeled as *real semantic change*, while those with higher similarity are labeled as *no significant drift*. The overlap in the middle region indicates that semantic drift is not strictly binary and requires both quantitative measures and interpretation.

Discussion and Analysis

The results demonstrate that semantic drift is not a uniform or binary phenomenon, but rather occurs along a spectrum with varying degrees of change. Two distinct patterns of evolution were observed. In cases such as “*viral*”, the word undergoes a semantic shift, where a newer meaning (internet popularity) becomes dominant over the original meaning (medical context). This is supported by the clear separation of clusters and a higher drift score. In contrast, words like “*mouse*” exhibit semantic expansion, where both traditional (animal) and modern (computer device) meanings coexist. The overlapping clusters and

lower drift score indicate that the original meaning has not disappeared but instead shares usage with a newer interpretation. These observations highlight that semantic drift often involves addition or dominance of meanings rather than complete replacement.

Another important insight is the role of data sources in influencing semantic change. Reddit data, representing informal and user-generated content, shows a faster emergence of new meanings compared to news data, which tends to preserve more formal and stable usage. This confirms that social media plays a significant role in accelerating language evolution. However, this also introduces challenges such as data imbalance, where the larger volume and variability of Reddit contexts may bias the results toward modern meanings. Additionally, clustering results reveal that meanings are not always cleanly separable; intermediate or overlapping regions suggest transitional phases where meanings gradually evolve rather than abruptly change.

The analysis also highlights limitations of purely quantitative metrics. While embedding similarity and drift scores provide a reliable signal of semantic difference, they cannot fully distinguish between true semantic change and contextual variation. This is evident from cases where words with relatively high drift scores were still classified as having “no significant drift.” To address this, the project incorporates an LLM-based explanation layer, which helps interpret whether the observed differences correspond to meaningful changes in usage. This combination of embedding-based detection and LLM-based interpretation provides a more balanced approach, ensuring both robustness and interpretability.

Overall, the project demonstrates that semantic drift detection requires a combination of quantitative modeling and qualitative reasoning. Embeddings and clustering effectively capture patterns in usage, while visualizations and LLM explanations help contextualize these patterns. The findings emphasize that language evolution is gradual, context-dependent, and influenced by social factors, reinforcing the need for adaptive NLP systems that can respond to ongoing changes in meaning.

Lessons Learned

- **Semantic drift is gradual, not binary**
Word meanings rarely change abruptly; instead, they evolve over time with overlapping phases where old and new meanings coexist.
- **Context is more important than the word itself**
Meaning is determined by how a word is used in sentences, not by the word in isolation. Extracting rich contextual examples is critical for accurate analysis.
- **Embedding-based methods are effective but not sufficient alone**
Cosine similarity and drift scores provide strong quantitative signals, but they cannot fully distinguish between true semantic change and contextual variation.

- **Clustering helps reveal hidden structure in meaning**
K-Means clustering was useful in identifying distinct semantic usages, but it also showed that meanings are often not cleanly separable and may overlap.
- **Social media accelerates language evolution**
Reddit data revealed faster emergence of new meanings compared to news data, highlighting the impact of informal, user-generated content on language change.
- **Data quality and balance significantly affect results**
Issues such as noisy text, misspellings, and uneven distribution between datasets can influence drift detection and need careful filtering.
- **LLMs improve interpretability but should not drive decisions**
Using LLMs for explanation adds clarity and human understanding, but relying on them for detection can introduce bias or overestimation of drift.
- **Precomputing and pipeline design improves efficiency**
Extracting candidate words and contexts in advance enables fast, flexible analysis without recomputing expensive steps.
- **Visualization is essential for understanding results**
Plots such as clustering diagrams and similarity vs drift graphs help reveal patterns that are not obvious from numerical metrics alone.
- **Semantic drift detection requires a hybrid approach**
Combining embeddings, clustering, visualization, and LLM-based explanations provides a more complete and reliable understanding than any single method alone.

Limitations

- **Dataset Imbalance**
The Reddit dataset contains significantly more data than the news corpus, which can bias the results toward modern meanings and exaggerate semantic drift.
- **Preprocessing Complexity and Time Cost**
Cleaning, structuring, and organizing large-scale textual data (especially Reddit) required substantial time and computational effort, making the pipeline resource-intensive.
- **LLM Output Reliability**
The LLM-generated explanations are not always accurate and may misinterpret contexts or overgeneralize meanings. Improving prompt design or fine-tuning is necessary for more reliable outputs.
- **Lack of Ground Truth**
There is no labeled dataset for semantic drift, making it difficult to objectively validate the correctness of detected changes.

- **Sensitivity to Thresholds**

The drift detection depends on similarity thresholds, which are manually chosen and may not generalize well across all words.

- **Clustering Limitations**

K-Means assumes a fixed number of clusters and may not capture more complex or overlapping semantic structures in language.

- **Context Quality Issues**

Noisy text, slang, misspellings, and inconsistent usage in Reddit data can affect embedding quality and drift measurement.

Future Work

Future work will focus on improving both the accuracy and applicability of the semantic drift detection pipeline. One key direction is to develop an adaptive thresholding mechanism that can automatically determine when meaningful semantic change has occurred, instead of relying on fixed similarity thresholds. Building on this, the pipeline can be extended to include a retraining trigger system, where the model monitors newly emerging words or significant shifts in existing words and signals when retraining is necessary. This is particularly important in real-world applications, as frequent retraining of large language models is computationally expensive, and a selective, data-driven approach can help reduce cost while maintaining performance.

Additionally, improvements can be made to the LLM explanation module by refining prompts or fine-tuning models to generate more accurate and reliable interpretations of semantic change. Future work may also explore temporal embedding models that capture continuous evolution over time rather than discrete early vs late comparisons. Enhancing clustering methods to better handle overlapping meanings, incorporating more balanced and diverse datasets, and extending the system to support real-time drift monitoring are also important directions. Finally, scaling the pipeline to larger vocabularies and applying it across multiple domains or languages would further strengthen its usefulness in understanding and adapting to evolving language patterns.

Team members contribution

Ullas Basavapatna Chandrashekar

- Responsible for data preparation and pipeline design
- Collected and prepared Reddit datasets from selected subreddits (AskReddit, YouTube, GenZ, GenAlpha, Decadeology) to capture modern language usage
- Integrated news dataset (Kaggle) to represent traditional meanings

- Implemented the data preprocessing pipeline, including:
 - Cleaning text (removing URLs, special characters, normalization)
 - Structuring Reddit data into monthly JSONL format for temporal analysis
 - Extracting and organizing word frequency timelines from news data
- Designed the overall project pipeline workflow
- Developed candidate word selection and drift score computation modules
- Contributed to presentation preparation and visualization of results

Vishrutha Ravi

- Responsible for semantic analysis and core modeling pipeline
- Implemented contextual embeddings using Sentence Transformers to capture word meaning based on usage
- Developed semantic clustering using K-Means to distinguish old vs new meanings
- Built visualization components (UMAP) to analyze and present semantic separation
- Designed and implemented the LLM-based explanation module for interpreting semantic changes
- Integrated embedding-based drift detection with LLM explanations
- Led analysis, evaluation, and report writing, including interpretation of results and insights